Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 38

Adjacency Matrices and Families of Graphs

Covers Chartrand & Zhang, Section 12.1

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

Three checks on any adjacency matrix

1. The diagonal is all 0.
2. It is symmetric.
3. Row i sums to $\deg v_i$, and all entries sum to $2m$.

If any check fails, you have made an error — or it is not an adjacency matrix.

Powers count walks

The (i, j) entry of A^k is the number of v_i–v_j walks of length k. In particular the diagonal of A^2 is the degree sequence.

Three mistakes to expect

1. **Confusing walks with paths.** A^k counts walks, which may repeat.
2. **Forgetting symmetry** when filling the matrix in.
3. **Reading the size as the entry sum.** That sum is $2m$.

Part II: Practice Problems

Section A — Building and reading

1. Let $V = \{v_1, v_2, v_3, v_4\}$ and $E = \{v_1v_2, v_1v_3, v_1v_4, v_2v_3\}$.

(a) Write the adjacency matrix A.

(b) Give the degree sequence and the size, reading both from A.

2. For the matrix A above, compute A^2 and use it to answer:

(a) How many walks of length 2 join v_2 and v_4?

(b) Why is the $(1, 1)$ entry of A^2 equal to 3?

Section B — Families

3. Give the order and size of each.

(a) K_8

(b) $K_{4,5}$

(c) C_9

(d) P_9

4.

(a) How many graphs are there on 5 labelled vertices?

(b) Is C_6 bipartite? Is C_7? Justify without finding a split.

Section C — Writing

5. In four to six complete sentences, explain why the adjacency matrix of a graph is always symmetric with a zero diagonal, and what each would mean if violated.